Model Card - Movie Rating Prediction Using ML and DL.

Model Details

- Project Developed by Frank Yan and Timing Yang at the University of Southern California, 2024.
- 3 Machine Learning (ML) models: Bayesian Ridge, Support Vector Regression, Random Forest Regression
- 4 Deep Learning (DL) Models: PCA-MLP, PCA-Attention, MLP, Attention.

Intended Use

- Intended to assist decision-makers in the movie industry, by predicting movie perceptions by its crew, cast, production, etc.
- Examples of usage such as choosing cast and deciding budget.
- Results should only be taken as suggestions, as the model cannot account for all features.

Factors

- Evaluation Factors: Genres, Keywords, Production Companies, Production Countries, Spoken Languages, Cast, Crew.
- Factors are chosen because almost all entries have those data, and their effects can be easily analyzed.

Metrics

- Mean Square Error and R^2 Score for Evaluation.

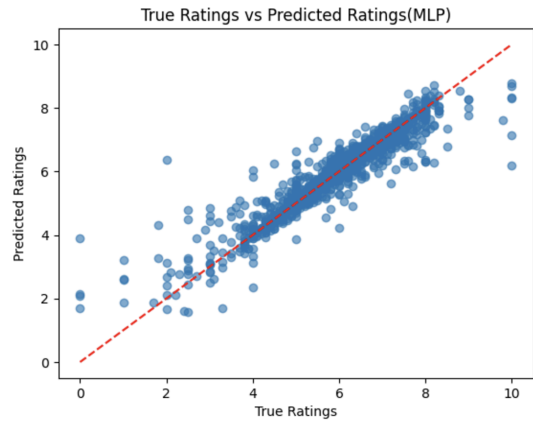
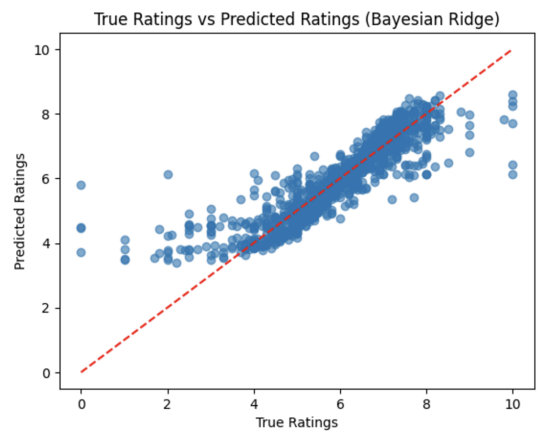
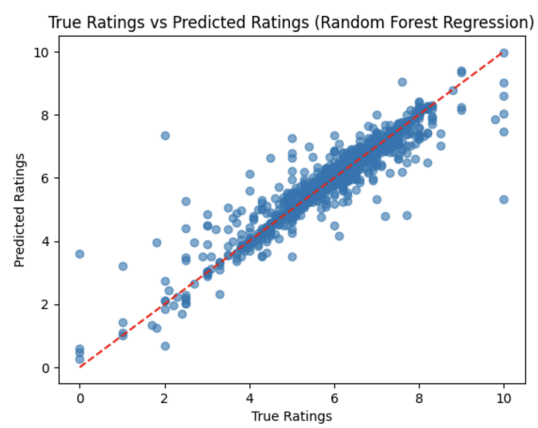
Training & Evaluation Data

- The Movie Dataset [1], Contains 50000 past movies.
- Training, testing splitting ratio of 0.7/0.3

Ethical Considerations

- Ratings reflect user demographics of IMDb-style platforms
- Popularity bias toward mainstream films
- Historical bias in cast/crew opportunities

Recommendations

- The ML models are the least complex models, followed by the PCA DL models, therefore those are the suggested models.
- If the movie dataset can be maintained and more trainable features can be made available, the performance of models can be improved significantly.

Quantitative Analysis